INCH-POUND

MIL-A-83212A(USAF)
AMENDMENT 2
27 February 1989
SUPERSEDING
AMENDMENT 1
6 April 1983

MILITARY SPECIFICATION

ALTIMETER, PRESSURE AAU-27/A

This amendment forms a part of MIL-A-83212A(USAF) dated 28 October 1976, and is approved for use within the Department of the Air Force and is available for use by all Department and Agencies of the Department of Defense.

PAGE 1

* 2.1 Delete "title and first sentence" and add: "Government documents."

* Add: "2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2)."

Under SPECIFICATION, Federal, add: "PPP-B-601 Boxes, Wood, Cleated Plywood."

Under SPECIFICATIONS, Military, the title of "MIL-L-25467" should be changed to read: "Lighting, Integral, Red, Aircraft Instrument, General Specification For".

Delete address on line 3 of beneficial comments block and add: "Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990.

PAGE 2

Under STANDARDS, Military, Delete" MIL-STD-143", "MIL-STD-794" and add: "MIL-STD-970 Standards and Specification, Order of Preference For The Selection Of" and "MIL-STD-2073-1 DOD Material Procedure For Development And Application Of Packaging Requirements." Change title "Publication" to "Military Handbook".

* Add: "(Copies of AFSC Design Handbooks are available from ASD/ENES Wright-Patterson AFB, OH 45433-6503)."

PAGE 3

* Delete the parenthetical source statement that follows the listing of Government specifications and standards; add: "(Unless otherwise indicated, copies of federal and military specifications and standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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* Add: "2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents which is cited in the solicitation (6.2).

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services)."

* Add: "2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (excpet for related associated detail specifications, specification sheets or MS standards), the text of this document take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

3.1 Delete paragraph and add new: "3.1 First article. When specified (see 6.5), a sample shall be subjected to first article inspection in accordance with 4.4.

PAGE 4

3.4.2.2 Delete paragraph and add new: "3.4.2.2 Mean time between failures (MTBF). The altimeter shall demonstrate a lower test MTBF (θ_1) of 1000 hours to a 90 percent confidence level."

PAGE 12

3.8.4.1 Middle column of table: Add: "Height" just above "Inch $\pm$ 0.015."

PAGE 14

* Add: "3.18 Recycled and reclaimed materials. Recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item."

* Add: "4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is any acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material."

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* Delete "4.2" "a", "b" and add new: "4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspections (see 4.4).

b. Quality conformance inspection (see 4.5)."

* Delete "4.3" and add new: "4.3 Inspection conditions. Unless otherwise specified all inspections shall be performed in accordance with the test conditions specified in 4.3.1 of this specification."

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* Delete "4.4" entirety and add new: "4.4 First article inspection."

* 4.4.1 Line 4 delete "procuring" and add: "contracting."

* 4.4.2 Delete "Preproduction tests" and add: "First article" in (2) two places.

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4.6.22.1 Delete paragraph and add: "4.6.22.1 First article. The first article reliability test shall be conducted in accordance with MIL-STD-781, using test conditions of paragraph 4.6.22.5 Reliability test conditions, herein which shall demonstrate an altimeter lower test MTBF (θ_1) of 1000 hours to a 90 percent confidence level. The altimer shall be pneumatically cycled from minimum to maximum altitude and return to minimum (one cycle) within approximately 15 minutes for a total of at least four cycles per hour. During the external vibration period (10 minutes of each hour), the cycling rate shall be increased to 20,000 fpm. Period B shall be approximately 4 hours in duration (see figure 4). During the cooling period, the power shall be turned off and the altitude input returned to sea level. The baro set knob shall be rotated once throughout the complete range each 24-hour period. In case of malfunction, the malfunction shall be recorded, the test interrupted while repairs are made, the test then continued from the point of interruption. At approximately 200-hour intervals, the altimeter shall be subjected to the tests specified in 4.6.2, 4.6.2.1, 4.6.2.2, 4.6.3, 4.6.4 4.6.4.1, 4.6.6.1, and 4.6.6.1.1. At the conclusion of this test, the altimeter shall meet all individual tests. If recalibration, adjustment, or repair is required to pass any tests, a failure shall be charged against this test. A 25-foot out-of-tolerance condition in scale error will be permitted provided rezeroing by use of the baro set system will correct the condition. This rezeroing shall not result in any out-of-tolerance condition. Scale erroor tests at all test temperatures shall be performed afer rezeroing. Rezeroing will be allowed only once per instrument per test. During all cycling, the vibrator shall be operating and any failure shall be charged as a malfunction against this test. The lighting system shall also be operating during cycling and upon completion of the final cycle, the altimeter shall meet the lighting tests specified in 4.6.7, 4.6.12, and 4.6.19. Failure to meet these tests shall be considered a malfunction. Time accured during operation of the altimeter in previous tests may be counted as part of the time required by this test. This test may be interrupted for normal weekend shutdown. Accept-reject criteria shall be in accordance with MIL-STD-781, test plan IIIC. If five or more failures occur, the altimeter will have failed the test."

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Add new figure 4, which forms page 7 of this amendment.

Page 28

4.6.22.2 Delete existing paragraph and add: "4.6.22.2 Production. The production reliability test shall be conducted in accordance with 4.6.22.1 except accept-reject criteria shall be in accordance with figure 5, herein."

Add new figure 5, which forms page 8 of this amendment.

4.6.22.3 Delete paragraph entirely and add: "4.6.22.3 Longevity. Two of the six first article reliability test samples shall be subjected to a longevity test in accordance with 4.6.22.3.1 and 4.6.22.3.2. and 4.6.22.3.3."

Add: "4.6.3.1 General. The equipment shall be tested at the specified test level for a period of time equivalent to the life requirement in this specification. The longevity test shall be 3000 operating hours for each sample. Test time accumulated on the equipment during a reliability test may be applied toward the longevity test when the same sample units are used."

Add: "4.6.22.3.2 Sample size. At least two (2) samples shall be subjected to the longevity test."

Add: "4.6.22.3.3 Evaluation of longevity test. No accept-reject criteria for this test is specified. The evaluation shall be limited to the analysis of all failures, identification of design and pattern failures, and the submission of recommended design changes to the contracting activity."

Add new paragraph as follows:

"4.6.22.5 Reliability test conditions. Reliability test conditions are as follows:

Temperature.....-54°C to +55°C (-65°F to 131°F)

Temperature Cycling.....Temperature cycling shall be time to stabilize at the high temperature, plus 2 hours.

Vibration.....2.2G ± 10% peak acceleration value at any nonresonant frequency between 20 and 60 Hz measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of operating time.

Equipment on-off cycling.....Equipment off during cooling cycle and on during heating cycle. (see figure 4)

Input voltage.....Nominal specified voltage +5 -2%."

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* 4.7 Line 2 delete "MIL-STD-794"and add: "MIL-STD-2073-1".

* Delete Section 5 in its entirety and add:

"5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Altimeter shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2 Drying. Altimeter shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each altimeter shall be packaged in quantity unit packs of one each in accordance with Method IIc of MIL-P-116. Each altimeter shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the altimeter when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each altimeter shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of altimeter shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Altimeter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Altimeter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable rules and regulations.

5.2.4 Industrial. The packaged altimeter shall be packed in accordance with ASTM D3951.

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5.3 Marking. In addition to any other marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordnace with MIL-STD-129."

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* 6.2 Delete "title and first sentence" and add: "Acquisition requirements. Acquisition documents must specify the following:". Add new "e. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1)."

6.3 Line 2, delete "STANAG 3319 and ASCC Air Standard 10/7" and add: NAT-STD-3319 and NAT-STD-3626 and ASCC Air Standard 10/7, 10/60, and 10/62."

* Delete "6.4" entirety.

Add new paragraph 6.5 as follows: "6.5 First article. When a first article is required, the item will be inspected and tested in accordance with Section 4 of this document. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracues in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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Preparing activity:
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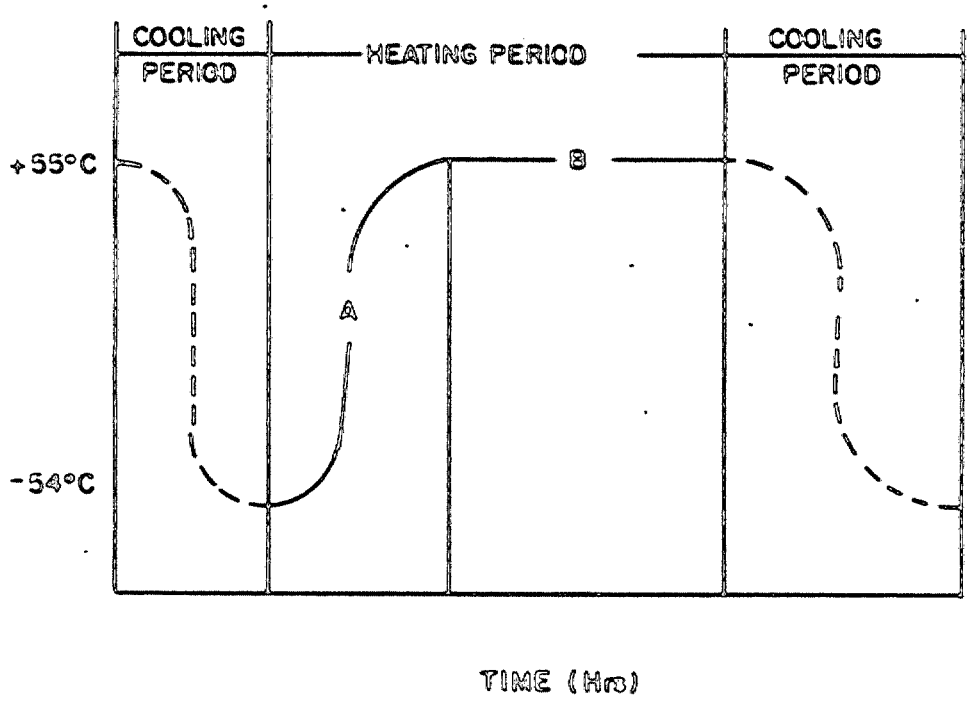
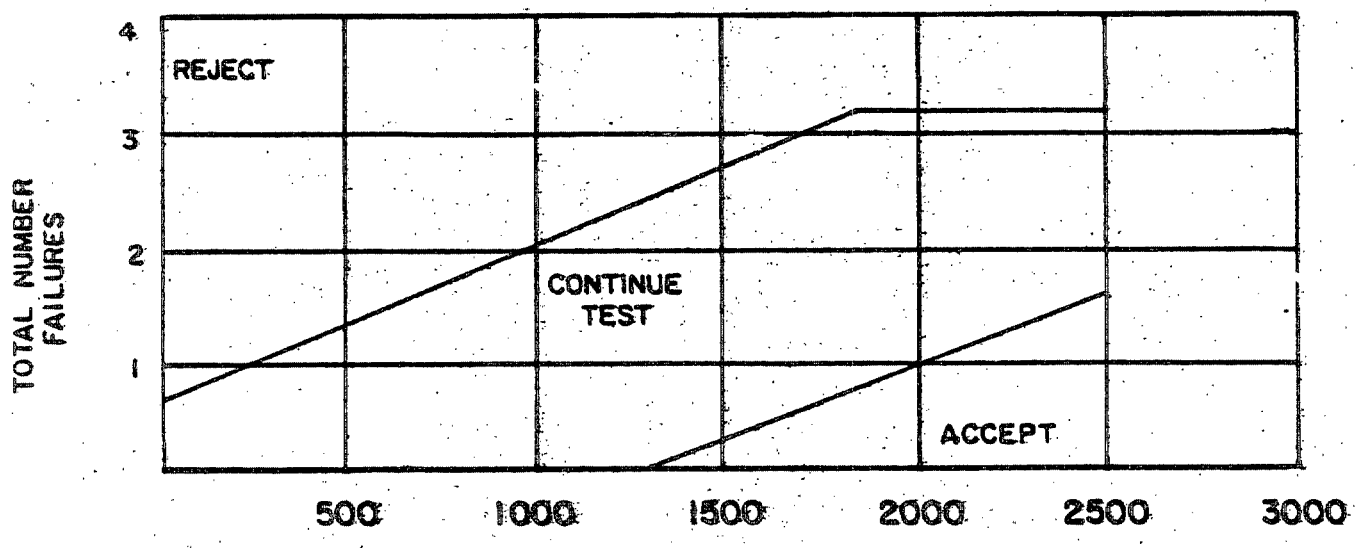


FIGURE 4. Equipment on-off cycling.

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NO. OF FAILURES	TOTAL TEST TIME (HRS) REJECT (EQUAL OR LESS)	ACCEPT (EQUAL OR MORE)
0	N/A	1100
1	80	1900
2	880	2500
3	1700	2500
4	2500	N/A

FIGURE 5. Accept-reject criteria.

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